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Synthesis of silver nanoparticles of some edible basidiomycetes mushroom fungi using response surface methodology and its potential biological application

R Madhanraj[1], M Eyini[2] and **P Balaji[1]***

[1]PG and Research Centre in Biotechnology,
MGR College, Hosur, Tamil Nadu, India

[2]Centre for Research and PG studies in Botany,
Thiagarajar College, Madurai, Tamil Nadu, India

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Abstract Metal Nanoparticles (*m*-NPs) are explored in recent years as an alternative approach to effectively kill drug resistant pathogenic microorganisms. Silver nanoparticles (AgNPs) are an important class of nanomaterial for a wide range of industrial and biomedical applications. Synthesis of silver nanoparticles through biological route is preferred over chemical route to promote green chemistry due to its environment-friendly and economic aspects. Mushrooms have been known for their nutritional and culinary values and used as medicines and tonics by humans for ages. In modern terms, they can be considered as functional foods which can provide health benefits beyond the traditional nutrients. The present study demonstrates an eco-friendly and low-cost method of biosynthesis of silver nanoparticles using basidiomycetes mushroom fungal strains *Lentinus tuberrigium*, *Lentinus psuedotenebrosus* and *Trametes sp.* The synthesized AgNPs were confirmed by the colour transformation and UV-visible spectroscopy. The appearance of dark brown colour and UV absorption spectra confirmed the synthesized silver nanoparticles. These biologically synthesized AgNPs were tested for antimicrobial, antioxidant, anti-diabetic and anti-hemolytic effects. The biosynthesized AgNPs showed significantly higher antimicrobial, antioxidant, anti-diabetic and anti-hemolytic activity. The present study explored the mushrooms which are efficient producers of AgNPs and could act as safe and cost-effective with potential biological applications.

Key Words Silver nanoparticles, Basidiomycetes mushroom, UV-visible spectroscopy, Antimicrobial, Antioxidant.

Introduction

The rapidly emerging field which has led to hopeful innovative applications in engineering and medical fields in terms of their economy, efficacy and economy is Nanotechnology. In the field of nano-drug delivery systems nanoparticles are attaining a great interest and also the cells of the body organs are not damaged during nano-drug delivery. Hence, there is a need to develop green chemistry methodologies in the synthesis for the nanomaterials. In this aspect, synthetic methods based on naturally occurring biomaterials are the different eco-friendly method (Ravishankar et al., 2011). The synthesis of gallant metal nanoparticles and their depiction attracts an growing interest in the field of nanotechnology because of their potential applications in various fields such as biotechnology, chemistry, physics and medicine (Jha et al., 2014). Nanosilver is the most protruding nanoproduct among several nanoproducts. AgNPs have been used for potential biological effects.

*balaji_paulraj@yahoo.com

Silver nanoparticles (AgNPs) are an imperative for a pervasive range of industrial and biomedical applications (Sangiliyandi et al., 2013). AgNPs are known to have antioxidant and antimicrobial properties (Mohamed et al., 2014).

Silver Nanoparticles (AgNPs) had become progressively popular as an antibiotic agent in textiles and wound dressings, medical devices, and appliances (Stensberg et al., 2011). Conversely, a low yield were given by physical methods and lethality to the environment were given by chemical methods, and also to prepare AgNPs with well-defined size it was greatly challenging, synthesis of AgNPs by means of enzymatic reduction, with better control over the size and shape of the NPs were given by biological methods (Wani and Ahmed, 2013; Wani et al., 2013). The progress of environmentally benevolent materials like fungi for the synthesis of AgNPs offers plentiful benefits of eco-friendliness and compatibility for biomedical and other pharmaceutical applications, as they do not use lethal chemicals for the synthesis protocol (Devika et al., 2012). Filamentous fungi are the improved agents for large-scale synthesis of nanoparticles in bioreactors and biomass production in contrast to bacteria and algae this is because fungal mycelial mat can effectively withstand various conditions in the bioreactors. Some fungi studied were found to be proficient of both intra-cellular and extra-cellular biosynthesis of AgNPs (Supriyo et al., 2014).

The use of fungi exclusively mushrooms in the synthesis of nanoparticles is a new accumulation and holds hopeful role in large scale nanoparticles production in less time. In laboratory and industrial level they are simpler to grow and the yield is also high (Manzoor-ul-Haq et al., 2015). In the fungus *Fusarium semitectum* synthesis of silver nanoparticles was reported and their medicinal applications of these silver nanoparticles have also been predicted (Mohan et al., 2014). The fungus kingdom comprises around 1.5 million species; though only 14,000 are basidiomycetes (Blackwell, 2011). On an industrial scale few of these genera are cultivated worldwide, including *Lentinula, Pleurotus*, *Agaricus*, *Agrocybe*, *Auricularia*, *Flammulina*, *Volvariella and Coprinus* (Li, 2012). For a variety of biotechnological applications mushrooms are used, particularly for the production of enzymes, dietary supplements, pharmaceutical compounds and food (Elisashvili, 2012); feed supplements (Santos et al., 2015). From various cellular components and secondary metabolites basidiomycete medicinal properties were originated that can be identified and isolated in the vegetative in the culture broth, mycelia or fruiting body (Elisashvili, 2012).

In the present consequence nonmaterials arose up as innovative antimicrobial agents owing to their high surface area to volume ratio and distinctive physical and chemical properties (Shivaraj et al., 2014). In the field of biology and medicine silver nanoparticles play a significant role and show a strong toxicity over microorganisms. Silver nanoparticles are reported to possess antimicrobial activities (Samberg et al., 2011). The extremely small size is the fact that the mechanism of antimicrobial property of nanoparticle lies with. Silver nanoparticles are hence a benign alternative to antibiotics (Sarmistha et al., 2011).

Edible mushrooms possess potent antioxidants which have been concluded by many studies and it was found that the antioxidant activity was exhibited by the crude ethanol extract of about 150 mushrooms. A

widespread exploration for traditional plant treatments for diabetes has resolved that predictable edible mushrooms are an ideal food for the dietetic preclusion of hyperglycemia (Sushila et al., 2012). Antioxidative constituents are now supposed to be potential protective agents also in contradiction of wood degrading organisms. Although natural antioxidants such as a-tocopherols and L-ascorbic acid are widely used, investigations are being carried out to discover more potent, safer antioxidants (Sablik et al., 2015).

In this work, nanoparticles were synthesized from mycelial extracts of the selected mushroom fungi (*Lentinus psuedotenebrosus, Trametes sp.* and *Lentinus tuberregium)* and in the culture filtrates. Antimicrobial activities, antioxidant activities and antidiabetic assays of selected mushroom culture filtrate and their nanoparticles were studied. This potentially can lead to novel therapeutic, imaging, and biomedical applications.

Experimental details

Basidiomycetes Fungi

The pure cultures of basidiomycetes mushroom fungal strains used in the present study were *Lentinus tuberrigium, Trametes sp and Lentinus psuedotenebrosus* and pure cultures were maintained in Centre for Research and PG studies in in Biotechnology, M.G.R. College, Hosur, Tamilnadu, India.

Silver nanoparticles synthesis

Synthesis of Silver nanoparticles

Silver nanoparticles were synthesized using aqueous solution (1mM) of silver nitrate. To the 95 ml of aqueous solution of 1 mM silver nitrate, 5 ml of mushroom broth/ extract was added for reduction into Ag^+ ions. To 9.5 ml of 10^{-3} M $AgNO_3$ aqueous solution 0.5 ml of nanoparticles broth/ extract was added in a boiling tube and heated on water bath at 75 °C for 60 min this was followed in a typical synthesis of silver (Ag). The color change from colorless to brown confirms the reduction of silver nitrate to silver ions. The formation of silver nanoparticles was also confirmed by spectrophotometric determination. The fully reduced solution was centrifuged at 5000 rpm for 30 min. The pellet obtained was redispersed in deionized water and the supernatant liquid was discarded. To wash off any absorbed substances on the surface of the silver nanoparticles, the centrifugation process was repeated two to three times (Veerasamy et al., 2010).

Fixation of different parameters

The temperature was maintained at 37°C, 45°C, 50°C, 60°C and 75°C respectively using water bath. The pH was maintained at 6, 7 and 8 respectively. The reaction was monitored from 15 min to 75 min at 15 min time interval. The reaction was monitored using different concentration of silver nitrate (0.5, 1, 1.5, 2 and 2.5 mM). The absorbance of the resulting solutions was measured spectrophotometrically.

Central composite design and response surface methodology

The experimental design was carried out by using “Stat-Ease Design-Expert” software (version 8.1, Stat-Ease Corporation, USA). The four independent factors were investigated at different levels like three (−2, 0, +2) (pH) and five (-4, −2, 0, +2, +4) (temperature, concentration of silver nitrate and time) respectively. The response Y (yield of PHB) was analysed by using a second order polynomial equation in four independent

variables and the data were fitted into the equation by multiple regression procedure.

The model equation for analysis is given below Eq.

$$Y = \beta 0 + \Sigma\beta\ iXi + \Sigma\ \beta\ iiX2\ i + \Sigma\ \beta\ ijXiXj$$

where Y is the predicted response, Xi, Xj represent the independent variables which influence the response variable Y, and ß0, ßi, ßii, and ßij represent the offset term, the linear coefficient, the quadratic coefficient and the interaction coefficient, respectively. "Design-Expert" 8.1 was used for regression and graphical analyses of the data obtained. Statistical analysis of the model was performed to evaluate the analysis of variance (ANOVA). The student's t-test permitted the checking of the statistical significance of the regression coefficient, and the Fischer's test determined the second-order model equation. The quality of the fit of the polynomial model equation was given by the coefficient of determination (R2). The data obtained by using the response surface regression procedure of the SAS statistical package (Version 8.1, SAS institute inc. NC. USA) were used to calculate the optimum concentration of the variables (Paulraj and Saravanan, 2012).

UV-V is Spectra analysis

UV-Vis spectral analysis was done by using UV-Visible spectrophotometer. UV-visible absorption spectrophotometer with a resolution of 5 nm between 200 and 800 nm possessing a scanning speed of 300 nm/min were used. UV-Vis spectrum of the reaction medium after diluting a small aliquot of the sample into deionized water were measured to monitor the reduction of pure Ag+ ions. One milliliter of the sample was pipetted into a test tube and diluted with 4 ml of deionized water and subsequently analyzed at room temperature (Shrivastava et al., 2007).

Antimicrobial Activity

Antimicrobial activity was measured by agar well diffusion assay. Fresh cultures of test bacterial strains were grown overnight in nutrient broth to get 10^5 CFU/ml. 0.1ml of each bacterial suspension was used to inoculate individual petridish containing nutrient agar medium with a sterile non-toxic cotton swab on a wooden applicator. Eight millimeter diameter 5 wells were punched in the agar plates. The mycelia (1g) of the selected mushroom fungi fresh biomass were homogenized in tris buffer (1 ml). The homogenate was centrifuged (6000 rpm for 10 minutes). Aliquots (25 µl, 50 µl, 75 µl, 100 µl and 125 µl) of the aqueous extracts were used to fill the wells. The plates were incubated at 28 ± 2°C overnight. Anti-bacterial activity was recorded by measuring the zone of growth inhibition around the well. The same procedure was followed for studying the antimicrobial effect of culture filtrate.

Antioxidant activity

Determination of total phenolic content

Ten microliters of aliquot of the extracts (2 mg/2 ml) were taken and made up to the volume of 1 ml with distilled water in the test tubes. Then 2.5 ml of sodium carbonate solution and 0.5 ml of folin-ciocalteu phenol reagent were added serially in each tube. The test tubes were placed dark for 40 min soon after vortexing the reaction mixture, and against the reagent blank the absorbance was recorded at 640nm. The results were stated as the gallic acid equivalents and the investigation was accomplished in triplicate (Siddhuraju and Becker, 2003).

Hydrogen peroxide radical scavenging activity

In phosphate buffer (0.2 M, pH 7.4), a solution of hydrogen peroxide (2 mmol/L) was prepared. Spectrophotometrically from absorption at 230nm hydrogen peroxide concentration was determined with molar absorbtivity 81 $(mol/l)^{-1}/cm$. Extracts (10 µl) were added to 3.4 ml of phosphate buffer together with hydrogen peroxide solution (0.6 ml). After 10 min against a blank solution containing phosphate buffer without hydrogen peroxide, the absorbance of hydrogen peroxide at 230 nm was determined. The percentage inhibition was calculated from $[(A_0-A_{1)/}A_0)]$ X100, where A_0 is the absorbance of the control (reaction mixture without extract) and $A_1$is the absorbance of the extract/ standard.

Phosphomolybdenum assay

To 1ml of reagent solution (0.6M sulphuric acid, 28mM sodium phosphate and 4mM ammonium molybdate) an aliquot of 100µl of sample solution (in 1mM dimethly sulphoxide) was combined in a 4ml vial. The vials were capped and incubated in a water bath at 95^0C for 90 min. At 695nm against a blank after the samples had cooled to room temperature the absorbance of the mixture was measured. The results reported (Ascorbic acid equivalent antioxidant activity) are mean values expressed as g of Ascorbic acid equivalents/100 g extract (Prieto et al., 1999).

Estimation of total flavonoid content

To 2 ml of distilled water and subsequently with 0.15 ml of a 5% $NaNO_2$ solution, 0.5 ml extract was mixed. After 6 min, addition of 0.15 ml of a 10% $AlCl_3$ solution and incubtion for 6 min were done, then 2 ml of 4% NaOH solution was added to the mixture. Distilled water was added immediately to bring the final volume of 5 ml, and allowed to stand for another 15 min after the mixture is thoroughly mixed. At 510 nm versus prepared water blank, the absorbance of the mixture was determined. Rutin was used as a standard compound for the quantification of total flavonoid. Per 100 gram of extract, all the values were stated as rutin (gram) equivalent (RE) (Zhishen et al., 1999).

Determination of reducing power

10-100µg of extract in 1ml of methanol was mixed with 5 ml of 0.2 M phosphate buffer (pH 6.6) and 5ml of 1.0% potassium ferric cyanide and the mixture was incubated at 50^oC for 20 min. After incubation 5ml of 10% trichloro acetic acid (TCA) was added to the mixture, which was then centrifuged at 1000rpm for 10 min. To 5ml of distilled water 5 ml of upper layer of supernatant was mixed and 0.5 ml of 0.1% ferric chloride. Then the absorbance was read spectro photometrically at 700 nm. A higher absorbance of the reaction mixture indicated greater reducing power (Siddhuraju et al., 2002).

Anti-hemolytic activity

By centrifugation the erythrocytes from cow blood were separated and with phosphate buffer (pH 7.4) were washed. With phosphate buffered saline the erythrocytes were then diluted to give 4% suspension. To 2ml of the erythrocyte suspension 500µg of extract/ml of saline buffer were added and with saline buffer the volume was made up to 5ml. At room temperature the mixture was incubated for 5 min and in saline buffer 0.5ml of H_2O_2 solution was added to induce the oxidative degradation of the membrane lipids. The concentration of H_2O_2 in the reaction mixture was adjusted to bring about 90%

hemolysis of blood cells after 240 min. after incubation the reaction mixture was centrifuged at 1500 rpm for 10 min and the extend of hemolysis was determined by measuring the absorbance at 540nm corresponding to hemoglobin liberation. The percentage of hemolysis was calculated from $[(A_0-A_{1)/}A_0)]$ X100, where A_0 is the absorbance of the control (reaction mixture without extract) and A_1 is the absorbance of the extract/ standard (Naim et al., 1976).

Anti-diabetic assay

Starch azure (2 mg) was suspended in 0.2 mL of 0.5M Tris-HCl buffer (pH 6.9) containing 0.01 M $CaCl_2$ (substrate solution). The tubes containing substrate solution were boiled for 5 min and then pre incubated at 37ºC for 5 min. Extract was dissolved in DMSO in order to obtain concentrations of 50, 250, 500 and 1000 µg/mL. Then, 0.2 mL of extract of particular concentration was added to the tube containing the substrate solution. In addition, 0.1 mL of porcine pancreatic amylase in Tris-HCl buffer (2 units/mL) was added to the tube containing the plant extract and substrate solution. The reaction was carried out at 37ºC for 10 min. The reaction was stopped by adding 0.5 mL of 50% acetic acid in each tube. The reaction mixture was centrifuged at 3000 rpm for 5 min at 4ºC. The absorbance of resulting supernatant was measured at 595 nm using spectrophotometer. Acarbose, a known α-amylase inhibitor was used as a standard drug (Hansawadi et al., 2000). The experiments were repeated thrice. The α-amylase inhibitory activity was calculated by using following formula:

$$\text{The } \alpha\text{ - amylase inhibitory activity} = \frac{\text{Test OD - Control OD}}{\text{Control OD}} \text{X } 100$$

The α-amylase inhibitory activities of crude extracts, nanoparticles and a carbose standard were calculated.

Results and Discussion

Synthesis of silver nanoparticles

As mushroom broth/ extracts were mixed into aqueous solution of the silver nitrate, it started to change the colour from watery to brown due to reduction of silver ion; which indicated the formation of silver nanoparticles.

Central composite design and response surface methodology

Central Composite Design is powerful method for screening significant factor in the presence study, 30 runs were carried out to investigate the optimization of synthesis of silver nanoparticles using four different factors *viz.*, temperature, pH, concentration of silver nitrate and time. The synthesis of silver nanoparticles varied from 0.16 to 1.48 grams/100ml of silver nitrate. Response surface methodology helps in evaluation of relationship between the dependent (silver nanoparticles) variable and independent variables.

Among these basidiomycetes mushrooms *Lentinus psuedotenebrosus* broth culture filtrate showed highest yield of silver nanoparticles. Among these parameters temperature vs. concentration and pH vs. concentration showed highly optimized (Fig. 1 and Fig. 2) and others parameters were moderately optimized.

UV-V is spectra analysis

In aqueous suspensions UV–Vis spectroscopy is used to examine shape and size-controlled nanoparticles. Fig. 3 shows the UV-Vis spectra recorded from the reaction medium of samples after heating the solution at 75ºC for 60 min. Absorption spectra of *Lentinus tuberregium, Lentinus psuedotenebrosus* and *Trametes sp.* silver

nanoparticles formed in the reaction media has absorbance peak at nm=422nm, A=1.52; nm=428nm, A=1.61 and nm=420nm, A=1.73 respectively and broadening of peak indicated that the particles are polydispersed.

Antimicrobial activity

Screening for antibacterial activity against microorganisms including bacteria such as *Bacillus sp.*, *Proteus sp.* and *Staphylococcus aureus* was carried out to determine whether the mushroom broth/ extracts, silver nanoparticles possessed any biological activity (Table 1).

Antioxidant activity

Table 2 shows the Total phenolic content, hydrogen peroxide scavenging activity (%), phosphomolybdenum assay and total flavonoid content of the culture filtrates of the selected mushroom fungi and synthesized nanoparticles.

Phenolic content of the extract

Phenolic compounds which were determined by Folin ciocalteau's method were reported as gallic acid equivalents by reference to standard curve ($Y = 0.0215 X$, $R^2 = 0.9887$). The scavenging effect of *Lentinus psuedotenebrosus* broth extracts has the highest value of 63.811 mg gallic acid equivalent/g of extract. The scavenging effect of *Lentinus tuberregium* nanoparticles has the highest value of 14.28 mg gallic acid equivalent/g of extract.

Hydrogen peroxide radical scavenging activity

The scavenging of hydrogen peroxide by phenolic compounds on the broth and their silver nano particles have high electron donating ability with maximum of *Lentinus tuberregium* 89.7%.

Phosphomolybdenum assay

Phosphomolybdenum compound was determined were reported as ascorbic acid equivalence by reference to standard curve ($Y= 0.187x$; $R^2= 0.9951$). The scavenging effect of *Lentinus tuberregium* broth extract has the highest value of 73.33 mg ascorbic acid equivalent/g of extract. The scavenging effect of *Trametes sp.* nanoparticles has the highest value of 318.89 mg ascorbic acid equivalent/g of extract.

Total flavonoid content

The scavenging effect of *Lentinus psuedotenebrosus* broth extract has the highest value of 200 mg rutin equivalent/g of extract. The scavenging effect of *Trametes sp.* nanoparticles has the highest value of 653.3 mg rutin equivalent/g of extract.

Table 3 shows the reducing power of the culture filtrates of the selected mushroom fungi and synthesized nanoparticles. *Lentinus psuedotenebrosus* broth extract found to be more effective on reducing power with optical density value of 0.51 at 700 nm, at 100 µg/ml. *Lentinus tuberregium* nanoparticles found to be more effective on reducing power with optical density value of 0.50 at 700 nm, at 100 µg/ml.

Anti-hemolytic activity

Table 4 shows anti- hemolytic activity of the culture filtrates of the selected mushroom fungi and synthesized nanoparticles. Lipid oxidation of cow blood erythrocytes membrane mediated by hydrogen peroxide induces membrane damage and subsequent hemolysis. A high significant efficiency in inhibiting radical induced red blood cell hemolysis was observed in silver nano particles of *Trametes sp.* with 97.39%. A high significant efficiency was observed in

silver nano particles of *Trametes sp.* extract with 97.39%.

Antidiabetic assay

Table 5 shows *invitro* porcine pancreatic amylase activity of mushroom broth and silver nanoparticles. The result of the study revealed that there is an increase in the % inhibition concentration with increase in the concentration of the sample. The α-amylase inhibitory activity varied among the extracts tested. The most potent inhibition appeared to be present in the crude extract of *Lentinus psuedotenebrosus* with 99.23%**.** They strongly inhibited the α-amylase activity of the concentration of 1000 µg/ ml. The nanoparticles of *Trametes sp.* showed the highest inhibitory effect with 97.69%.

Conclusion

The growth conditions have been efficiently optimized for basidiomycetes mushroom fungi after preliminary screening. 19th day culture was found efficient and was used for nanoparticles synthesis. Silver nanoparticles of respective genera have been efficiently synthesised using central composite design in response surface methodology. The efficient nanoparticle synthesis was observed in *Lentinus psuedotenebrosus* (1.48 grams/100ml), with optimum parameters temperature - 50□C, pH – 8, time - 45 min, concentration- 1.5 mM silver nitrate. On screening of synthesized nanoparticles for biological activities, AgNP synthesized from *Lentinus psuedotenebrosus* showed efficient antimicrobial activity. The scavenging effect of *Lentinus tuberregium* nanoparticles has the highest value of 14.28 mg gallic acid equivalent/g of extract and has the maximum scavenging of hydrogen peroxide by phenolic compounds (89.7%). Also *Lentinus tuberregium* nanoparticles found to be more effective on reducing power with optical density value of 0.50 at 700 nm, at 100 µg/ml. The scavenging effect of *Trametes sp.* nanoparticles has the highest value of 318.89 mg ascorbic acid equivalent/g of extract and 653.3 mg rutin equivalent/g of extract. Also high significant efficiency of anti-hemolytic activity was observed in silver nano particles of *Trametes sp.* extract with 97.39% and showed the highest inhibitory effect with 97.69% which strongly inhibited the α-amylase activity of the concentration of 1000 µg/ ml. The present study reveals the production of silver nanoparticles from edible basidiomycetes mushroom fungal strains *Lentinus tuberrigium, Lentinus psuedotenebrosus* and *Trametes sp.*, and their potent antimicrobial, antioxidant, anti-diabetic and anti-hemolytic effects. The AgNPs from these medicinally and nutritionally important mushrooms could be effective, safe and cost effective.

Conflict of interest

We declare that there is no conflict of interest.

Table 1: The antimicrobial activity of the culture filtrates and synthesized nanoparticles

Test organism	Concentration (µl)	Zone of inhibition diameter (cm)					
		Name of the Sample					
		Lentinus tuberregium		*Lentinus psuedotenebrosus*		*Trametes sp.*	
		Crude	Nanoparticles	Crude	Nanoparticles	Crude	Nanoparticles
Bacillus sp.	25	1.9	1.4	2.0	1.6	1.2	1.3
	50	2.1	1.5	2.0	1.8	1.4	1.4
	75	2.1	1.7	2.1	2.0	1.6	1.5
	100	2.3	1.8	2.1	1.9	1.9	1.6
	125	2.2	1.9	2.2	2.0	2.0	1.8
Proteus sp.	25	1.3	-	1.1	1.1	-	-
	50	1.7	-	1.3	1.2	1.1	-
	75	1.6	1.0	1.4	1.4	1.1	1.0
	100	1.5	1.0	1.5	1.3	1.4	1.1
	125	1.8	1.1	1.6	1.4	1.5	1.1
Staphylococcus aureus	25	1.6	1.5	1.7	1.6	1.3	1.4
	50	1.9	1.8	1.8	1.7	1.5	1.4
	75	1.9	1.6	1.9	1.7	1.7	1.6
	100	2.0	1.9	2.0	1.8	1.8	1.5
	125	2.0	2.0	2.0	1.9	2.0	1.6

Table 2: Total phenolic content, Hydrogen peroxide scavenging activity (%), Phosphomolybdenum assay and Total flavonoid content of the culture filtrates of the selected mushroom fungi and synthesized nanoparticles

Name of the extract	Total phenolic compounds		Hydrogen peroxide scavenging activity (%)		Phosphomolybdenum assay		Total Flavonoid content	
	O.D value	Scavenging effect (mg GAE/1g)	O.D value	Scavenging effect	O.D value	Scavenging effect (mg AAE/1g)	O.D value	Scavenging effect (mg RE/1g)
Lentinus tuberregium crude	0.067	31.91	0.107	89.70	0.132	73.33	0.07	46.67
Lentinus psuedotenebrosus crude	0.134	63.81	0.109	89.51	0.097	53.89	0.30	200.0
Trametes sp. crude	0.033	15.71	0.110	89.41	0.019	10.56	0.06	40.00
Lentinus tuberregium nanoparticles	0.024	14.29	0.124	88.07	0.022	12.22	0.55	366.67
Lentinus psuedotenebrosus nanoparticles	0.021	10.00	0.125	87.96	0.025	13.89	0.36	240.0
Trametes sp. nanoparticles	0.009	4.286	0.123	87.77	0.574	318.89	0.98	653.3

Table 3: Reducing power of the culture filtrates of the selected mushroom fungi and synthesized nanoparticles

Concentration (µg/ml)	OD Values at 700nm					
	Lentinus tuberregium crude		*Lentinus psuedotenebrosus* crude		*Trametes sp.* crude	
	crude	nanoparticles	crude	nanoparticles	crude	nanoparticles
Control	1.904					
10	0.15	0.08	0.21	0.06	0.07	0.07
20	0.18	0.12	0.23	0.09	0.09	0.08
30	0.20	0.17	0.25	0.11	0.12	0.10
40	0.24	0.21	0.27	0.14	0.15	0.11
50	0.27	0.26	0.29	0.16	0.17	0.12
60	0.31	0.30	0.31	0.18	0.20	0.13
70	0.34	0.34	0.35	0.21	0.23	0.15
80	0.36	0.39	0.39	0.25	0.28	0.16
90	0.39	0.45	0.45	0.28	0.32	0.17
100	0.41	0.50	0.51	0.31	0.38	0.18

Table 4: Anti- hemolytic activity of the culture filtrates of the selected mushroom fungi and synthesized nanoparticles

S. No	Name of the extract	Concentration (µg/ml)	Optical density	Anti-hemolytic activity
1.	*Lentinus tuberregium* crude	500	0.91	20.87
2.	*Lentinus psuedotenebrosus* crude		0.97	15.65
3.	*Trametes sp.* crude		0.84	26.97
4.	*Lentinus tuberregium* nanoparticles		0.06	94.78
5.	*Lentinus psuedotenebrosus* nanoparticles		0.07	93.91
6.	*Trametes sp.* nanoparticles		0.03	97.39

Control OD =1.15

Table 5: *Invitro* Porcine pancreatic amylase activity

	Concentration (µg/ml)	OD value at 595 nm					
		Lentinus tuberregium		*Lentinus psuedotenebrosus*		*Trametes sp.*	
		O. D	% Inhibition	O. D	% Inhibition	O. D	% Inhibition
Mushroom Broth	50	0.22	83.08	0.26	80.00	0.25	80.77
	250	0.20	84.60	0.24	81.54	0.21	83.85
	500	0.05	96.15	0.07	94.62	0.15	88.46
	1000	0.02	98.46	0.01	99.23	0.02	98.46
Silver Nanoparticles	50	0.22	83.08	0.20	84.62	0.15	88.46
	250	0.13	90.00	0.15	88.46	0.12	90.78
	500	0.09	93.08	0.12	90.78	0.08	93.85
	1000	0.04	96.92	0.06	95.38	0.03	97.69

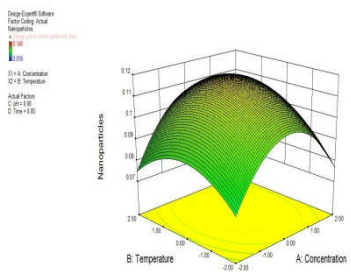

Fig. 1: Model graph of *Lentinus psuedotenebrosus* broth – Temperature vs Concentration

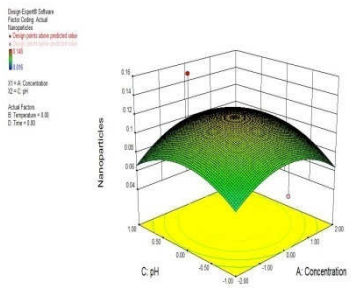

Fig. 2: Model graph of *Lentinus psuedotenebrosus* broth – pH vs Concentration

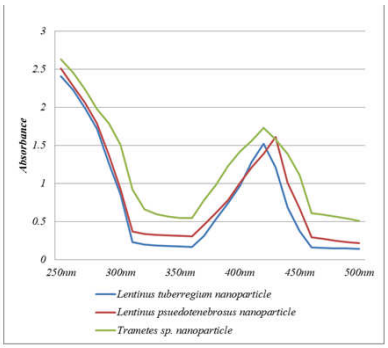

Fig. 3: UV-Vis spectra analysis of synthesized nanoparticles from selected basidiomycetes mushroom fungi

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